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Undergraduate Mathematics I

AN INTRODUCTION TO CALCULUS, ALGEBRA, ANALYSIS, AND THEIR
APPLICATIONS.

UNIVERSITY OF WESTERN AUSTRALIA

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Linear Algebra

Foundations

LINEAR ALGEBRA, at least at this stage, is the study of linear functions, their behaviour and interactions, where they live in space, and how they can be represented with matrices and determinants¹. The *linear* in linear function refers to the fact that any variables, x, y, z , etc, are never raised to any power or multiplied together, only ever multiplied by scalars² or added together. Geometrically this means that linear functions never curve. (fig. 1)

Linear functions generally take the form:

$$y = mx + b$$

Where m and b are numbers, and x and y are variables. In this form, m represents the *slope* of the line, how steep it appears on the plot, and b represents the y -intercept, the point at which the line crosses the vertical axis.

You can also define a line by the points it intersects (fig. 2). You only need two points to figure out the equation representing a line. Imagine a line traveling through the points $(-1, -1)$ and $(1, 2)$. The slope of this line, its *rise over run*, would be $y_2 - y_1$ divided by $x_2 - x_1$ (The height gained divided by the distance travelled), and the y -intercept would be $y_{1,2} - mx_{1,2}$, obtained by rearranging the form we want the equation to take.

$$\begin{aligned} y &= \left(\frac{y_2 - y_1}{x_2 - x_1} \right) x + \left(y_1 - \left(\frac{y_2 - y_1}{x_2 - x_1} \right) x_1 \right) \\ \Rightarrow y &= \left(\frac{(2) - (-1)}{(1) - (-1)} \right) x + \left((-1) - \left(\frac{(2) - (-1)}{(1) - (-1)} \right) (-1) \right) \\ &\Rightarrow y = \left(\frac{3}{2} \right) x + \left(\frac{1}{2} \right) \end{aligned}$$

The middle line looks very complicated, but only due to my verbosity. They are just numbers.

¹ Sheldon Axler. *Linear Algebra Done Right*. Springer, 3rd edition, 2015

² Scalars are numbers, real or complex. It soon becomes obvious why they are called scalars.

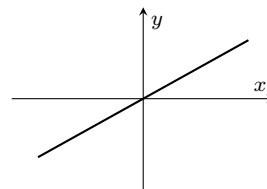


Figure 1: A linear function.

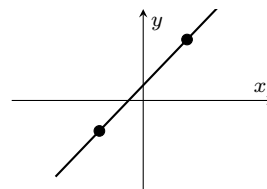


Figure 2: $(-1, -1)$ and $(1, 2)$ and $y = 1.5x + 0.5$

Vectors

VECTORS ARE QUANTITIES WITH A MAGNITUDE AND A DIRECTION. They are used to represent things like force or momentum. It does not make much sense, for example, just to say "the ball experiences a force of 5N", we must add where the force is directed, 5N *down*. Vectors are usually notated with an arrow above their name, $\vec{\text{vector}}$.

Vectors are usually written down just as a set of points, $\vec{u} = (2, 2)$. This translates to "the vector $\mathbf{u}$ starts at the origin $(0, 0)$ and ends at the point $(2, 2)$." From fig. 3 we can see that this vector forms a right triangle with base length 2, height 2 and, from the Pythagorean theorem, hypotenuse $\sqrt{2^2 + 2^2} \approx 2.83$. The angle from the x axis, θ is then $\cos^{-1}\left(\frac{2}{2.83}\right) = 45^\circ$.

If this vector, $\vec{u} = (2, 2)$, were being used to represent force, for example, the force would have a magnitude of 2.83N and a direction of 45° .

In three dimensions its slightly more complicated by much the same. The vector $\vec{r} = (2, 2, 2)$ starts at the origin, goes to the point $(x, y, z) = (2, 2, 2)$ and stops. We now need two angles and its length to fully describe it, the angle ϕ is the angle the projection of the vector on the $x - y$ plane makes with the x axis, and the angle θ is the angle the vector makes with the z axis.

To calculate ϕ , you ignore the z coordinate and pretend you just have a vector in 2 dimensions. Now you can figure out its angle with the x axis like we did above. Calculating θ is much the same, ignore the x coordinate, and now you just have a 2D vector again! (fig. 4) The length of this vector $\vec{r}$ is just the Pythagorean theorem in 3 dimensions, $|\vec{r}| = \sqrt{x^2 + y^2 + z^2} = \sqrt{2^2 + 2^2 + 2^2} \approx 3.46$.

We can generalise this concept to as many dimensions as we want. We say a vector is an *ordered n -tuple* of numbers, which just means a list, n items long³.

$$\vec{v} = (v_1, v_2, \dots, v_{n-1}, v_n) \text{ where } v_{1-n} \text{ are numbers.}$$

Vector Addition

VECTORS CAN BE ADDED by adding up their members. This produces another vector that, geometrically, has an angle from the x axis inbetween the angles of the two input vectors, with total length equal

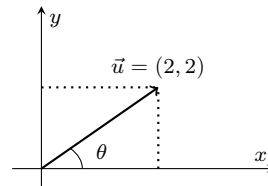


Figure 3: Vectors can be defined both by the point they end on, and their length and angle from an axis.

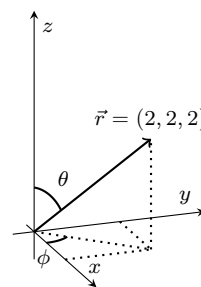


Figure 4: Vectors in 3D require 2 angles and a length.

³ In 2D, $n = 2$. In 3D, $n = 3$. A 1D vector is just a number.

to the sum of the lengths of each input vector (fig. 5).

$$\overrightarrow{u+v} = \vec{u} + \vec{v} = (u_1 + v_1, u_2 + v_2, \dots, u_n + v_n)$$

Scalar Multiplication

VECTORS ARE SCALED by multiplying them with a number. This produces a vector that points in the same direction but is longer by whichever *scaling factor* you multiplied by. This is achieved by distributing the number to all the members of the vector.

$$\lambda \vec{u} = (\lambda u_1, \lambda u_2, \dots, \lambda u_n) \text{ for some } \lambda \in \mathbb{R}$$

Vector Multiplication

VECTORS CAN BE MULTIPLIED in two ways; one that produces a number, and one that produces another vector. The *dot* product is the sum of the product of the respective members of two vectors of the same length.

$$\vec{u} \cdot \vec{v} = (u_1 \times v_1) + (u_2 \times v_2) + \dots + (u_n \times v_n)$$

It represents how much of one vector *goes* in the direction of another. This is particularly useful if one of the vectors you are taking the dot product with is a unit vector⁴ since it tells you how much of a vector goes in the unit vector's direction.

For example, if you have some vector representing force, $\vec{F}$, and you take its dot product with $\hat{i}$ you get the amount of force in the x direction, its x -component.

THE CROSS PRODUCT of two vectors produces another vector. Geometrically, this new vector is perpendicular to the plane containing the input vectors. We say that the cross product of two vectors $\vec{u}$ and $\vec{v}$, $\vec{u} \times \vec{v}$ is *normal* to the surface containing them (fig. 6). If you hold your right hand out in front of you in the shape of a gun, leave your pointer finger in the direction of $\vec{u}$ and bend your middle finger to the direction of $\vec{v}$, your pointer and middle fingers are two vectors that form a plane. Your thumb pointing up or down is normal to that plane. If your pointer was $\vec{u}$ and your middle finger $\vec{v}$ then your thumb would be their cross product, $\vec{u} \times \vec{v}$ (fig. 7).

That is, in fact, the definition of the cross product:

$$\vec{u} \times \vec{v} = |u| \cdot |v| \sin(\theta) \vec{n}$$

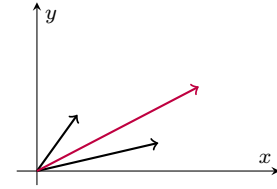


Figure 5: $\vec{u} = (1, 2)$, $\vec{v} = (3, 1)$ and $\vec{u} + \vec{v} = (4, 3)$

⁴ Unit vectors are vectors that travel along an axis and have length 1, e.g. $(1, 0)$ and $(0, 1)$ are the unit vectors of the $x-y$ plane. Unit vectors are denoted by a little hat. Traditionally, $\hat{i}$, $\hat{j}$, and $\hat{k}$ represent the unit vectors in the x , y , and z directions, respectively.

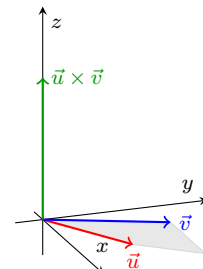


Figure 6: The cross product $\vec{u} \times \vec{v}$ is normal to the plane containing $\vec{u}$ and $\vec{v}$.

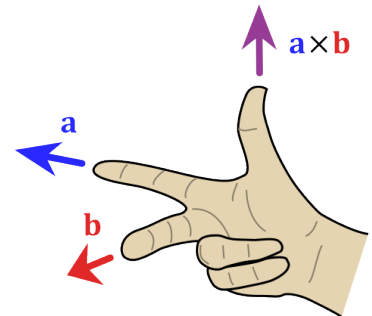


Figure 7: The right hand rule for cross products.

Where $|u|$ is the magnitude, or length, of $\vec{u}$, θ is the angle between $\vec{u}$ and $\vec{v}$, and $\vec{n}$ is the unit normal vector to the plane containing $\vec{u}$ and $\vec{v}$, determined via the right hand rule.

THE CROSS PRODUCT SATISFIES THE FOLLOWING RULES, OR AXIOMS.

1. The cross product is **anticommutative**.

$$\vec{u} \times \vec{v} = -(\vec{v} \times \vec{u})$$

Going back to the right hand rule from earlier, if you put your left hand out in front of you and try to make the same shape with your pointer and middle fingers as is on your right hand, the only way to do this is to have your thumb pointing down.

2. The cross product is **distributive over addition**.

$$\vec{u} \times (\vec{v} + \vec{r}) = \vec{u} \times \vec{v} + \vec{u} \times \vec{r}$$

3. Scalars can be factored out.

$$(\lambda \cdot \vec{u}) \times \vec{v} = \lambda(\vec{u} \times \vec{v})$$

4. A vector *dotted* with a cross product of itself is 0.

$$\vec{u} \cdot (\vec{u} \times \vec{v}) = 0$$

Recall from the definition of a dot product that it represents the amount of one vector that goes in the direction of another. If a cross product is perpendicular to both its input vectors then none of the input vector will *go* in the direction of the cross product, so by definition their dot product must be 0.

The standard unit vectors $\hat{i}, \hat{j}, \hat{k}$ satisfy the following equalities:

$$\hat{i} \times \hat{j} = \hat{k}$$

$$\hat{j} \times \hat{k} = \hat{i}$$

$$\hat{k} \times \hat{i} = \hat{j}$$

And by anticommutativity:

$$\hat{j} \times \hat{i} = -\hat{k}$$

$$\hat{k} \times \hat{j} = -\hat{i}$$

$$\hat{i} \times \hat{k} = -\hat{j}$$

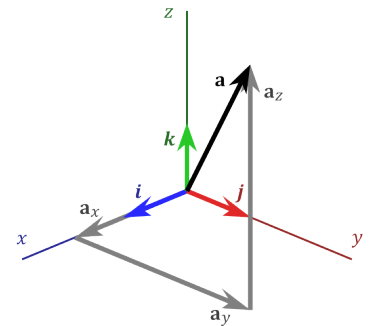


Figure 8: The unit vectors, $\hat{i}, \hat{j}, \hat{k}$ and the components of $\vec{a}$: $\vec{a}_x, \vec{a}_y, \vec{a}_z$

This implies that:

$$\hat{i} \times \hat{i} = \hat{j} \times \hat{j} = \hat{k} \times \hat{k} = \mathbf{0}$$

Since we also know that any vector $\vec{a}$ can be represented by $a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ (fig. 8). This is enough to derive⁵:

$$\begin{aligned} \vec{a} \times \vec{b} &= (a_1\hat{i} + a_2\hat{j} + a_3\hat{k}) \times (b_1\hat{i} + b_2\hat{j} + b_3\hat{k}) = \\ &= a_1b_1(\hat{i} \times \hat{i}) + a_1b_2(\hat{i} \times \hat{j}) + a_1b_3(\hat{i} \times \hat{k}) + \\ &+ a_2b_1(\hat{j} \times \hat{i}) + a_2b_2(\hat{j} \times \hat{j}) + a_2b_3(\hat{j} \times \hat{k}) + \\ &+ a_3b_1(\hat{k} \times \hat{i}) + a_3b_2(\hat{k} \times \hat{j}) + a_3b_3(\hat{k} \times \hat{k}) \\ &= a_1b_2(\hat{k}) - a_1b_3(\hat{j}) - a_2b_1(\hat{k}) + a_2b_3(\hat{i}) + a_3b_1(\hat{j}) - a_3b_2(\hat{i}) \\ &= (a_2b_3 - a_3b_2)\hat{i} + (a_3b_1 - a_1b_3)\hat{j} + (a_1b_2 - a_2b_1)\hat{k} \end{aligned}$$

⁵ Much simpler methods will be discussed later, however it is important to understand the derivation from the axioms.

VECTORS CAN BE MULTIPLIED AND ADDED IN THESE WAYS TO FORM A VECTOR SPACE, which will be defined properly later. Vectors and vector spaces are principle objects of study in linear algebra. Collections of vectors are called *matrices*.

Matrices

MATRICES ARE ORDERED ARRAYS OF NUMBERS. An $m \times n$ matrix has m rows and n columns.

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ a_{31} & a_{32} & \cdots & a_{3n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

Matrix Addition

MATRICES CAN BE ADDED if they are the same size and square. To add matrices simply add each element to the corresponding element in the other matrix.

$$\begin{bmatrix} a & c \\ b & d \end{bmatrix} + \begin{bmatrix} e & g \\ f & h \end{bmatrix} = \begin{bmatrix} a+e & c+g \\ b+f & d+h \end{bmatrix}$$

Matrix Scalar Multiplication

Like vectors, scalars can just be distributed to each element in a matrix.

$$\lambda \cdot \begin{bmatrix} a & c \\ b & d \end{bmatrix} = \begin{bmatrix} \lambda a & \lambda c \\ \lambda b & \lambda d \end{bmatrix}$$

Matrix Multiplication

TO MULTIPLY TWO MATRICES, you take the dot product of the rows of the first matrix with the columns of the second and *distribute* the results to the new matrix; i.e. first row $\cdot$ first column goes in the $(1, 1)$ position and first row $\cdot$ second column goes in the $(1, 2)$ position and so on. Thus the number of rows and columns must be equal; i.e. you can only multiply $m \times n$ matrices with $n \times p$ matrices. The result of a matrix $\mathbf{A}$ times a matrix $\mathbf{B}$ is called the matrix product $\mathbf{AB}$.

$$\begin{bmatrix} a_1 & a_3 & a_5 \\ a_2 & a_4 & a_6 \end{bmatrix} \times \begin{bmatrix} b_1 & b_4 \\ b_2 & b_5 \\ b_3 & b_6 \end{bmatrix} =$$

$$\begin{bmatrix} (a_1b_1 + a_3b_2 + a_5b_3) & (a_1b_4 + a_3b_5 + a_5b_6) \\ (a_2b_1 + a_4b_2 + a_6b_3) & (a_2b_4 + a_4b_5 + a_6b_6) \end{bmatrix}$$

MATRIX MULTIPLICATION SATISFIES THE FOLLOWING AXIOMS:

1. Matrix multiplication is non-commutative⁶.

If $\mathbf{A}$ and $\mathbf{B}$ are matrices of size $m \times n$ and $p \times q$ respectively then the matrix product $\mathbf{AB}$ is only defined if $n = p$ and $\mathbf{BA}$ only defined if $m = q$. If $n = p \neq m = q$ then both products are defined but not the same size so can't be equal and so the only case when both products are defined and could be equal is when $m = n = p = q$; i.e both matrices are square and the same size. Even in this case $\mathbf{AB}$ generally does not equal $\mathbf{BA}$. For example:

$$\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \times \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

but:

$$\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \times \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

Only when one of the matrices is the identity matrix⁷, or both are square diagonal matrices⁸, is $\mathbf{AB} = \mathbf{BA}$.

⁶ Recall an operation is commutative if given two elements $\mathbf{A}$ and $\mathbf{B}$ such that the product $\mathbf{AB}$ is defined, then $\mathbf{BA}$ is also defined and $\mathbf{AB} = \mathbf{BA}$

⁷ Defined below.

⁸ A matrix with all zeroes other than the main top-left to bottom-right diagonal.

2. Matrix multiplication is distributive over addition.

Given 4 matrices, $\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}$, and $\mathbf{D}$ with sizes $m \times n$, $n \times p$, $n \times p$ and $p \times q$, one has both left and right distributivity:

$$\mathbf{A}(\mathbf{B} + \mathbf{C}) = \mathbf{AB} + \mathbf{AC}$$

$$(\mathbf{B} + \mathbf{C})\mathbf{D} = \mathbf{BD} + \mathbf{CD}$$

3. Matrix multiplication is associative **iff** given 3 matrices $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$, the number of columns in $\mathbf{A}$ equals the number of rows in $\mathbf{B}$, and the number of columns in $\mathbf{B}$ equals the number of rows in $\mathbf{C}$.
Iff that is the case:

$$(\mathbf{AB})\mathbf{C} = \mathbf{A}(\mathbf{BC})$$

4. If the product $\mathbf{AB}$ is defined and $c \in \mathbb{R}$ then:

$$c(\mathbf{AB}) = (c\mathbf{A})\mathbf{B} \text{ and } (\mathbf{AB})c = \mathbf{A}(\mathbf{B}c)$$

The Identity Matrix

THE IDENTITY MATRIX is the special matrix $\mathbf{I}$, with the property that any other matrix $\mathbf{A}$ multiplied by it is unchanged. This property is known in mathematics as the *multiplicative identity*.

$$\mathbf{AI} = \mathbf{IA} = \mathbf{A}$$

It is a square diagonal matrix with its main diagonal all ones:

$$\begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{bmatrix}$$

Its size can be whatever is required for the multiplicative identity to hold.

Summary

The principle objects of study in linear algebra⁹ are linear functions, or functions of several variables where no variable is raised to any power, the vector, which is an *ordered n-tuple* of numbers usually representing some quantity with a magnitude and direction, the way vectors interact to form *spaces* through addition and multiplication, matrices, which are collections of vectors in an array, and the way matrices interact with each other. We will explore how the properties briefly explained here are useful for solving problems and see what structures follow from these few simple rules.

⁹ In this first pass.

Systems of Linear Equations

THE SYSTEMATIC SOLVING of systems of equations is a necessary skill for anyone wishing to study linear algebra. Many problems in this field can be reduced to a system of linear equations, and so solving them algorithmically is crucial.

A LINEAR EQUATION is an equation of the form:

$$x + 2y = 4$$

A *system* of linear equations, then, is a set of linear equations considered together:

$$x + 2y = 4$$

$$x - y = 1$$

Which is a system of two equations in two unknowns (fig. 9). You can have any number of equations in any number of unknowns but for there to be a unique solution, you at least need to have as many equations as you have unknowns. When we have n unknowns, we generally write linear equations as:

$$a_1x_1 + a_2x_2 + a_3x_3 + \cdots + a_nx_n = m \mid a_{1-n}, m \in \mathbb{R}, n \in \mathbb{N}.$$

Solutions to Systems of Linear Equations

A SOLUTION to a system of linear equations is any set of points $(x_1, x_2, \cdots, x_n)$ which satisfy every equation in the system. The system presented above has a unique solution at $(2, 1)$, but this is not always the case. Consider the simple system (fig. 10):

$$y - x = 0$$

$$y - x = 2$$

This is the same as the above $y = mx + b$. In this case $y = -\frac{1}{2}x + 2$.

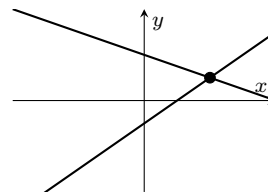


Figure 9: $x + 2y = 4$, $x - y = 1$ and their solution, $(2, 1)$.

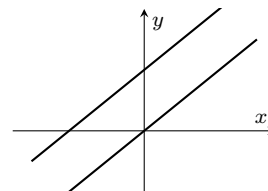


Figure 10: The lines $y - x = 0$ and $y - x = 2$ are parallel and thus have no solution.

The equations in this system are parallel and will never meet, thus there *can't* be a solution.

The system below doesn't have a unique solution for a very different reason.

$$\begin{aligned}y - x &= 2 \\ 2y - 2x &= 4\end{aligned}$$

Notice, we can divide both sides of the second equation by 2, leaving us with the first equation. These two lines are the same and so there are infinitely many solutions, every point on the line $y - x = 2$.

Now, imagine you knew three equations in two unknowns (fig. 11):

$$\begin{aligned}2x - y &= 1 \\ x + y &= 2 \\ x - 2y &= -2\end{aligned}$$

There is no solution of points $(x, y) = (a, b)$ that satisfies every equation, so the system has no solution. Just because you know more equations than unknowns, that does not guarantee that a solution exists. Can you conceive of a way to make this system have a solution? ¹⁰

That is the case for *every* system of linear equations, they either have no solutions, a unique solution or infinitely many solutions. The intuition for each is that the lines never cross, all cross at the same point, or are they same line. If a solution exists, we say the system is *consistent*. If one does not exist, we say it is *inconsistent*.

IN THREE DIMENSIONS linear equations form a plane. Consider the equation $x + y + z = 1$ (fig. 12). It is the plane of all points z that satisfy $z = 1 - x - y$. It is still possible to reason geometrically about linear equations in three dimensions, but beyond that it is impossible. We therefore need to develop *algebraic* methods of understanding these systems, not just geometric.

Imagine you only know two equations in three unknowns:

$$\begin{aligned}x + y + z &= 1 \\ x - y - z &= 1\end{aligned}$$

Each of these equations defines a plane and since those planes are clearly not parallel, they will meet in a line. Every point along that line will be a solution to this system.

One way of describing a system like this is to choose one of the variables to be free. If we chose z to be free, the first equation becomes

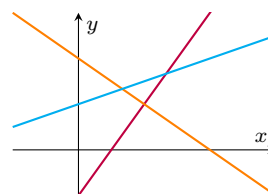


Figure 11: The plots of $2x - y = 1$, $x + y = 2$ and $x - 2y = -2$ never meet a single point.

¹⁰ Shift the blue line down a bit?

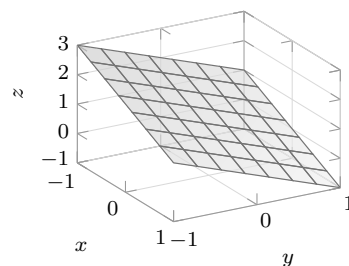


Figure 12: The plane $x + y + z = 1$.

$z = 1 - x - y$. We can then solve for the set of solutions by substituting this equation into the other:

$$x - y - (1 - x - y) = 1$$

$$2x = 2$$

$x = 1$ can now be used to solve for y .

$$1 + y + z = 1$$

$$y = -z$$

So a full solution to this system would be:

$$S = \{(1, -z, z) \mid z \in \mathbb{R}\}$$

Which means for any real number z , say $z = 3$, there is a solution $(1, -z, z) = (1, -3, 3)$ that satisfies both equations in the system.

Algorithmically Solving Systems of Linear Equations

AN ALGORITHM is a set of instructions for solving a specific problem that works every time. In the case of systems of linear equations, the algorithm is called *Gauss-Jordan Elimination* and involves altering the system in some way to make the solutions trivially obvious.

Building the Augmented Matrix

If you know m equations in n unknowns with $n, m \in \mathbb{N}$ and $a_n, k_n \in \mathbb{R}$:

$$a_1x_1 + a_2x_2 + a_3x_3 + \cdots + a_nx_n = k_1$$

$$b_1x_1 + b_2x_2 + b_3x_3 + \cdots + b_nx_n = k_2$$

$$c_1x_1 + c_2x_2 + c_3x_3 + \cdots + c_nx_n = k_3$$

$$\vdots$$

$$m_1x_1 + m_2x_2 + m_3x_3 + \cdots + m_nx_n = k_m$$

Then, following from the rules of matrix multiplication discussed in Chapter 1, you can build three matrices, $\mathbf{A}$, $\mathbf{x}$, and $\mathbf{B}$ such that:

$$\mathbf{Ax} = \mathbf{B}$$

Where $\mathbf{A}$ is a $m \times n$ matrix containing all the coefficients $a_{1 \rightarrow n} \rightarrow m_{1 \rightarrow n}$, $\mathbf{x}$ is a $n \times 1$ matrix containing all the variables $x_{1 \rightarrow n}$ and $\mathbf{B}$ is

a $m \times 1$ matrix containing all the scalars $k_{1 \rightarrow m}$:

$$\begin{bmatrix} a_1 & a_2 & \cdots & a_n \\ b_1 & b_2 & \cdots & b_n \\ \vdots & \vdots & \ddots & \vdots \\ m_1 & m_2 & \cdots & m_n \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} k_1 \\ k_2 \\ \vdots \\ k_m \end{bmatrix}$$

This notation is quite a lot to write out every time you want to solve a system of linear equations. Generally, when solving systems of linear equations by hand we build an *augmented matrix* with the coefficients on the left, and answers on the right; with the vertical line representing the " $\times \mathbf{x} =$ " statement:

$$\left[\begin{array}{cccc|c} a_1 & a_2 & \cdots & a_n & k_1 \\ b_1 & b_2 & \cdots & b_n & k_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ m_1 & m_2 & \cdots & m_n & k_m \end{array} \right]$$

It's done in this way because it makes the arithmetic of the following sections much easier to compute.

Elementary Row Operations

ELEMENTARY ROW OPERATIONS are operations applied to the equations in a system of linear equations to transform them into a simpler, but equivalent, system¹¹. This is done because the resultant system is trivially easy to solve. Gauss-Jordan elimination, then, is just a method for applying these operations that results in the same trivial system every time.

¹¹ Equivalent meaning they have the same solution.

THEY ARE:

1. Interchanging the order of equations.
2. Multiplying an equation by a scalar.
3. Adding a scalar multiple of one equation to another.

Theorem 1. *Let S be a system of linear equations and T be the resultant set of linear equations after applying arbitrary elementary row operations to S . The solution set of S equals the solution set of T .*

Proof: You should not need any convincing that the first two row operations do not alter the solution of the system. The third operation, however, requires some attention.

LET S be the set of linear equations $E_1 \rightarrow E_n$:

$$S = \{E_1, E_2, E_3, \dots, E_n\}$$

LET E_i and $E_j \in S$:

$$\begin{aligned} E_i &= a_{1i}x_1 + a_{2i}x_2 + \dots + a_{ni}x_n \\ E_j &= a_{1j}x_1 + a_{2j}x_2 + \dots + a_{nj}x_n \end{aligned}$$

LET T be the set of linear equations that results from the addition of $k \cdot E_j$ to E_i :

$$T = \{E_1, E_2, \dots, E'_i, E_j, \dots, E_n\}$$

THEN the only difference between S and T is E'_i :

$$E'_i = (a_{1i} + ka_{1j})x_1 + (a_{2i} + ka_{2j})x_2 + \dots + (a_{ni} + ka_{nj})x_n = E_i + kE_j$$

LET s be a solution to S :

$$s = (s_1, s_2, \dots, s_n) \mid s_i \in \mathbb{R}$$

THEN check if s is a solution to T :

$$\begin{aligned} E'_i &= (a_{1i} + ka_{1j})s_1 + (a_{2i} + ka_{2j})s_2 + \dots + (a_{ni} + ka_{nj})s_n \\ &= a_{1i}s_1 + ka_{1j}s_1 + a_{2i}s_2 + ka_{2j}s_2 + \dots + a_{ni}s_n + ka_{nj}s_n \\ &\text{which when grouped:} \\ &= (a_{1i}s_1 + a_{2i}s_2 + \dots + a_{ni}s_n) + k(a_{1j}s_1 + a_{2j}s_2 + \dots + a_{nj}s_n) \\ &= E_i + kE_j \end{aligned}$$

SO any solution to S is a solution to T but we still need to check that any solution to T is a solution to S .

IF a vector exists that satisfies all equations in T then it necessarily satisfies $E_i + kE_j$ and E_j . Therefore it also satisfies E_i since:

$$E_i + kE_j + (-kE_j) = E_i$$

THUS any solution to T is also a solution to S and the theorem holds.

□

Gauss-Jordan Elimination

GAUSS-JORDAN ELIMINATION is an algorithm for applying elementary row operations to augmented matrices¹² to transform them into a trivial-to-solve system.

¹² The system doesn't *need* to be represented as an augmented matrix, it just makes it easier.

REDUCED ROW ECHELON FORM is the final result of Gauss-Jordan Elimination. An augmented matrix is in reduced row echelon form if:

1. Any rows of all zeroes occur in the last row(s), and,
2. The first non-zero entry¹³ of each row is *strictly to the right* of the first non-zero entry of the row above, and,
3. The first non-zero entry of each row is equal to one.

¹³ The first non-zero entry is called the leading entry.

IN WORDS, Gauss-Jordan Elimination is carried out as follows: Let $\mathbf{A}$ be an $m \times n$ matrix. At each stage in the algorithm, a particular entry of the matrix, called the *pivot point* is being processed, and the steps of the algorithm depend on the value of this pivot.

1. Let the *pivot point* be the top-left $(1, 1)$ entry of the matrix. If this entry is 0, then interchange it with the row whose first entry is closest to zero. To make the process easier, if this entry is not 1, but there is a row with first entry equal to 1, interchange those two rows.
2. Divide the first row by the value of the first entry such that the first entry becomes 1.
3. Add and subtract as necessary the first row to other rows such that the first entry of every other row becomes 0.
4. Move the pivot point down and to the right 1 unit, i.e. to the $(2, 2)$ position of the matrix. If this entry is 0, then interchange rows other than the first row, such that this entry is non-zero.
5. Divide the second row by the value of its second entry, such that it becomes 1.
6. Repeat steps 3-5 with respect to the new shifted pivot point each time until the matrix is as close to the identity matrix as possible.

This algorithm laid out in words is quite difficult to understand, so

here is an example: The system (fig. 13):

$$\begin{aligned} 2x - y &= 1 \\ x + y &= 2 \\ x - 2y &= -1 \end{aligned}$$

Is represented by the augmented matrix:

$$\left[\begin{array}{cc|c} 2 & -1 & 1 \\ 1 & 1 & 2 \\ 1 & -2 & -1 \end{array} \right]$$

The first pivot point is the top left entry:

$$\left[\begin{array}{cc|c} \boxed{2} & -1 & 1 \\ 1 & 1 & 2 \\ 1 & -2 & -1 \end{array} \right]$$

Interchange rows 1 and 2 so that the top left entry is 1.¹⁴

$$\left[\begin{array}{cc|c} \boxed{1} & 1 & 2 \\ 2 & -1 & 1 \\ 1 & -2 & -1 \end{array} \right] \begin{array}{l} R_1 \iff R_2 \\ R_2 \iff R_1 \end{array}$$

Add or subtract the necessary multiples of R_1 such that the first entries of subsequent rows become 0 and shift the pivot down and to the right.

$$\left[\begin{array}{cc|c} 1 & 1 & 2 \\ 0 & \boxed{-3} & -3 \\ 0 & -3 & -3 \end{array} \right] \begin{array}{l} R_2 \iff R_2 - 2R_1 \\ R_3 \iff R_3 - R_1 \end{array}$$

Here we can just divide the entry by -3 since interchanging rows will have no effect.

$$\left[\begin{array}{cc|c} 1 & 1 & 2 \\ 0 & \boxed{1} & 1 \\ 0 & -3 & -3 \end{array} \right] R_2 \iff R_2 \div -3$$

Now we add or subtract the necessary multiples of R_2 such that the 2nd entry of every other row becomes 0.

$$\left[\begin{array}{cc|c} 1 & 0 & 1 \\ 0 & \boxed{1} & 1 \\ 0 & 0 & 0 \end{array} \right] \begin{array}{l} R_1 \iff R_1 - R_2 \\ R_3 \iff R_3 + 3R_2 \end{array}$$

We don't need to shift the pivot point down again since the last row is all 0. Now the solution can just be read off the matrix, $(x, y) = (1, 1)$.

Geometrically this has the effect of *rotating* each line about the solution (fig. 14).

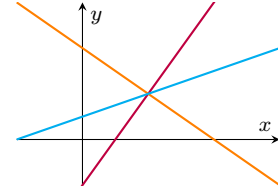


Figure 13: The plots of $2x - y = 1$, $x + y = 2$ and $x - 2y = -1$.

¹⁴ The $\iff$ is used to indicate when elementary row operations have been applied. $R_1 \iff R_2$ means R_1 becomes R_2 . $R_2 \iff R_2 - 2R_1$ means R_2 becomes itself minus 2 times R_1 .

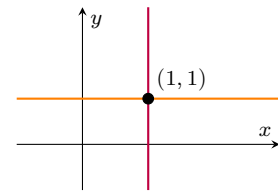


Figure 14: Elementary row operations rotate lines about their solution.

A MORE COMPLICATED EXAMPLE. The system represented by the augmented matrix:

$$\left[\begin{array}{cccc|c} 1 & 0 & 1 & 3 & 2 \\ 2 & 1 & 2 & 0 & 0 \\ -1 & 3 & 4 & 0 & 1 \end{array} \right]$$

Following the steps of Gauss-Jordan Elimination, we chose $(1, 1)$ to be our pivot:

$$\left[\begin{array}{cccc|c} \boxed{1} & 0 & 1 & 3 & 2 \\ 2 & 1 & 2 & 0 & 0 \\ -1 & 3 & 4 & 0 & 1 \end{array} \right]$$

We then add multiples of the first row to the second and third rows such that their first entries all become 0.

$$\left[\begin{array}{cccc|c} \boxed{1} & 0 & 1 & 3 & 2 \\ 0 & 1 & 0 & -6 & -4 \\ 0 & 3 & 5 & 3 & 3 \end{array} \right] \quad \begin{array}{l} R_2 \iff R_2 - 2R_1 \\ R_3 \iff R_3 + R_1 \end{array}$$

We then shift the pivot down and to the right.

$$\left[\begin{array}{cccc|c} 1 & 0 & 1 & 3 & 2 \\ 0 & \boxed{1} & 0 & -6 & -4 \\ 0 & 3 & 5 & 3 & 3 \end{array} \right]$$

And add multiples of the second row to the third row so that its second entry becomes 0.

$$\left[\begin{array}{cccc|c} 1 & 0 & 1 & 3 & 2 \\ 0 & \boxed{1} & 0 & -6 & -4 \\ 0 & 0 & 5 & 21 & 15 \end{array} \right] \quad R_3 \iff R_3 - 3R_2$$

Shift the pivot once again:

$$\left[\begin{array}{cccc|c} 1 & 0 & 1 & 3 & 2 \\ 0 & 1 & 0 & -6 & -4 \\ 0 & 0 & \boxed{5} & 21 & 15 \end{array} \right]$$

And finally divide the third row by 5 and add multiples of it to the first row such that its third entry becomes 0.

$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & -1.2 & -2.2 \\ 0 & 1 & 0 & -6 & -4 \\ 0 & 0 & 1 & 4.2 & 3 \end{array} \right] \quad \begin{array}{l} R_1 \iff R_1 - R_3 \\ R_3 \iff R_3 \div 5 \end{array}$$

This matrix is in reduced row echelon form and can't be reduced any further. This system has three *basic*¹⁵ variables, x_1, x_2 , and x_3 , and one *non-basic* variable x_4 . When writing the parametric solution to a system with free variables, like we have here, it is crucial to always

¹⁵ A variable is *basic* if its column contains the leading entry of a row, otherwise its *non-basic*. Consistent systems with more unknowns than equations will always have at least one non-basic variable. All non-basic variables are free variables, and all basic variables are linear combinations of constants and non-basic variables.

chose the non-basic variables as free. This leaves us with the following linear equations:

$$\begin{aligned}x_1 - 1.2x_4 &= -2.2 && \implies x_1 = -2.2 + 1.2x_4 \\x_2 - 6x_4 &= -4 && \implies x_2 = -4 + 6x_4 \\x_3 + 4.2x_4 &= 3 && \implies x_3 = 3 - 4.2x_4\end{aligned}$$

And so the solution will be:

$$S = \{(-2.2 + 1.2x_4, -4 + 6x_4, 3 - 4.2x_4, x_4) \mid x_4 \in \mathbb{R}\}$$

Reasoning about Systems of Linear Equations

HOW CAN WE TELL how many solutions a system has, or if it's consistent at all, just by looking at its augmented matrix?

If, after completing Gauss-Jordan elimination, one of the leading entries is on the right of the augmenting line, the system is inconsistent. If a system is consistent and all variables are basic variables, then there is a unique solution. If the system is consistent and there are any free variables, the system has infinitely many solutions.

EXAMPLES

$$\begin{array}{ccc}\left[\begin{array}{ccc|c} 1 & 0 & 1 & 1 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 3 \end{array} \right] & \left[\begin{array}{ccc|c} \boxed{1} & 0 & 1 & 1 \\ 0 & \boxed{1} & 2 & 1 \\ 0 & 0 & \boxed{2} & 3 \end{array} \right] & \left[\begin{array}{ccc|c} 1 & 0 & 1 & \boxed{1} & 2 \\ 0 & 1 & 2 & \boxed{1} & 1 \\ 0 & 0 & 2 & \boxed{3} & 4 \end{array} \right] \\ \text{Inconsistent} & \text{Unique Solution} & \text{Infinite Solutions}\end{array}$$

The first is inconsistent since there is a leading entry after the augmenting line. This says $0x + 0y + 0z = 3$ which does not make any sense, so there can't be any solutions. The second system doesn't have any leading entries after the augmenting line and has as many equations as unknowns, so all the variables are basic and there is a unique solution $(x, y, z) = (-0.5, -2, 1.5)$. The third has more unknowns than equations and is consistent, and so one of the variables must be non-basic and free. Therefore, there is an infinite solution set $S = \{(0.5x_4, 2x_4 - 3, 2 - 1.5x_4, x_4) \mid x_4 \in \mathbb{R}\}$.

Summary

Solving systems of linear equations quickly and easily is crucial to the study of linear algebra, since so many of the problems we will face will boil down to a system of linear equations. We can solve systems of linear equations by first finding the matrix equation $\mathbf{Ax}=\mathbf{B}$, and

the subsequent augmented matrix $\mathbf{A}|\mathbf{B}$; then identifying if the system is consistent by identifying the basic and non-basic variables, and applying elementary row operations with the method described by Gauss-Jordan elimination. In Chapter 3, we will finally define properly the idea of vector spaces, and use the tools developed in this chapter to solve various problems related to them.

Vector Spaces and Subspaces

THE CONCEPT OF A VECTOR SPACE is fundamental to the study of linear algebra.

PART VII

Analysis

PART VIII

Quantum Mechanics

Bibliography

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